

ORCAS ISLAND, WA, USA

WMO: 722208

Lat: 48.708N

Lon: 122.911W

Elev: 10

StdP: 101.21

Time Zone: -8.00 (NAP)

Period: 04-19

WBAN: 04224

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.9	-2.0	-13.0	1.2	-1.4	-10.1	1.6	0.9	8.8	8.6	8.1	7.7	2.8	50	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	23.0	16.9	22.4	16.7	21.2	16.3	18.0	22.0	17.3	21.1	16.7	20.2	2.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.6	19.1	15.2	10.8	18.5	14.9	10.6	18.3	50.6	22.2	48.6	21.2	46.5	20.3	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.6	6.8	5.7	DB	-6.1	26.6	1.8	1.4	-7.4	27.6	-8.4	28.5	-9.4	29.3	-10.7	30.3	(4)
(5)			WB	-7.6	19.4	2.1	1.0	-9.1	20.1	-10.3	20.7	-11.4	21.2	-13.0	21.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	5.5	5.1	7.0	9.1	12.4	14.8	17.0	17.0	14.3	10.5	7.5	5.0	(6)	
	DBStd	5.02	3.15	3.08	2.46	2.18	2.30	1.98	1.78	1.58	1.96	2.25	3.17	3.24	(7)	
	HDD10.0	687	143	137	95	40	6	0	0	0	0	21	86	158	(8)	
	HDD18.3	2888	399	369	350	276	184	108	48	48	122	243	326	414	(9)	
	CDD10.0	857	3	1	3	14	81	144	218	216	128	36	10	2	(10)	
	CDD18.3	16	0	0	0	0	0	2	8	6	1	0	0	0	(11)	
	CDH23.3	46	0	0	0	0	1	4	25	16	1	0	0	0	(12)	
	CDH26.7	2	0	0	0	0	0	0	2	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.4	2.7	2.5	2.4	2.3	2.2	2.5	2.5	2.3	1.8	2.1	2.7	2.6	(14)	
(15) Precipitation	PrecAvg	658	95	54	56	43	41	32	16	28	33	76	108	86	(15)	
	PrecMax	807	134	100	102	76	76	63	35	91	102	194	209	140	(16)	
	PrecMin	454	28	3	25	3	12	6	2	2	0	21	31	41	(17)	
	PrecStd	92	26	23	19	17	20	15	10	24	24	38	45	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	12.9	14.2	17.4	22.1	23.8	26.0	25.0	22.6	17.8	17.0	13.9	(19)	
		MCWB	10.4	10.3	10.0	12.2	15.2	16.9	18.3	17.3	17.1	14.3	13.1	10.8	(20)	
	2%	DB	12.4	12.0	12.8	15.1	19.0	22.1	23.7	23.0	20.2	16.2	13.9	12.3	(21)	
		MCWB	10.1	9.7	9.5	10.9	14.1	16.1	17.2	17.1	16.0	13.0	11.8	10.3	(22)	
	5%	DB	11.1	10.1	12.2	13.7	17.7	19.9	22.4	22.3	18.9	14.9	12.7	10.9	(23)	
		MCWB	9.6	8.3	9.2	10.0	13.2	15.2	16.7	16.9	15.5	12.3	10.9	8.7	(24)	
	10%	DB	9.9	9.0	11.0	12.6	17.0	18.7	21.2	21.0	17.7	13.9	12.1	9.0	(25)	
		MCWB	8.5	7.6	8.5	9.4	12.9	14.8	16.2	16.6	14.5	11.7	10.6	7.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.2	11.8	13.0	15.7	17.8	19.5	18.8	18.1	15.3	13.9	12.1	(27)	
		MCDB	12.7	12.4	13.5	16.6	20.0	22.5	24.1	23.0	21.1	17.1	14.6	13.2	(28)	
	2%	WB	10.7	9.7	10.4	11.6	14.6	16.5	18.0	17.9	16.7	14.0	12.5	10.3	(29)	
		MCDB	11.8	10.9	12.0	14.1	18.1	20.2	22.4	21.8	19.2	15.6	13.7	11.7	(30)	
	5%	WB	9.4	8.7	9.5	10.8	13.8	15.7	17.1	17.4	15.8	13.1	11.4	8.9	(31)	
		MCDB	10.5	9.4	11.1	12.9	17.2	19.1	21.3	21.0	18.1	14.5	12.7	9.9	(32)	
	10%	WB	8.1	7.9	8.7	10.1	13.0	14.8	16.5	16.8	15.1	12.3	10.3	7.9	(33)	
		MCDB	9.0	8.6	10.1	12.2	16.2	18.0	20.4	20.2	17.3	13.6	11.5	8.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.8	5.7	6.2	7.2	7.2	8.3	8.1	6.6	5.4	4.6	4.0	(35)	
		MCDBR	5.4	5.8	6.8	7.5	8.7	9.2	9.9	9.7	8.3	6.2	5.5	5.1	(36)	
	5% WB	MCWBR	3.3	3.8	4.1	4.3	4.9	4.8	5.0	5.0	4.7	3.9	3.6	3.5	(37)	
		MCWBR	4.9	5.1	6.2	6.7	8.5	8.4	9.4	9.0	7.5	5.7	4.9	4.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.319	0.315	0.330	0.357	0.379	0.356	0.351	0.357	0.342	0.332	0.321	0.312	(40)	
		taud	2.520	2.556	2.526	2.443	2.393	2.477	2.519	2.503	2.545	2.559	2.529	2.499	(41)	
	Ebn at Noon		754	852	890	894	882	905	904	883	867	816	743	710	(42)	
		Edh at Noon	61	72	87	104	114	105	100	97	84	71	60	55	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.91	1.78	2.76	4.21	5.42	5.90	6.48	5.47	3.87	2.12	1.08	0.72	(44)	
	RadStd		0.13	0.25	0.34	0.41	0.44	0.60	0.48	0.38	0.32	0.25	0.18	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page